OBAIDA GUL

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CORE SKILLS

- UI/UX
- Communication Skills
- C++
- Front end development
- Python
- NLP
- Machine Learning
- Computer Vision
- Speech Processing

PROFESSIONAL SUMMARY

Innovative and analytical developer with expertise in UI/UX design, Front-End Development, and Machine Learning (Computer Vision, NLP, and Speech Processing). Proficient in Python and C++, with a strong ability to solve complex problems and deliver high-performance solutions. Skilled in effective communication and cross-functional collaboration, ensuring seamless project execution under tight deadlines. Passionate about leveraging cutting-edge technologies to build scalable, user-centric applications while maintaining robust code quality.

LANGUAGES

- English
- Urdu
- Saraiki

EDUCATION

- **Al-Suffah College, Mianwali**
  Matric (Science **)**
- **Punjab Group of college** ,Mianwali
  FSC (Pre- engineering)
- **National University of Technology** Islamabad, Pakistan
  BS in Artificial Intelligence

PROJECTS

- **Real-Time Gesture Detection System for Smart Home Control**
  TensorFlow, Keras, MediaPipe, and OpenCV were used to construct a real-time hand gesture recognition system. Hand landmarks were detected and tracked using MediaPipe, and motions were identified by a CNN model constructed using TensorFlow/Keras. OpenCV was integrated to process live video feeds and carry out actions in real time. Using Arduino, certain motions were linked to home automation functions like turning fans or lights on and off, mimicking a touchless smart house management system.

- **Vehicle Counting Using Object Detection**
  Built a vehicle counting system with TensorFlow, Keras, and OpenCV utilizing the COCO-pretrained Mask R-CNN model. Utilized object detection to detect vehicles in real-time traffic video and applied a counting algorithm by following objects crossing a virtual line. Facilitated traffic flow analysis and congestion monitoring for smart city use.

- **Fall Prevention and Obstacle Detection Wheelchair Robot**
  Developed an intelligent robotic wheelchair prototype that used TensorFlow, Numpy, and Pandas for fall detection and prevention. Applied TensorFlow to execute real-time detection models and obtained data through ultrasonic sensors. Data processing using Pandas ensured proper classification of fall risk situations. The system was able to detect objects and modify navigation to prevent collisions or harmful movements.

- **AI-Based X-ray Disease Classification System**
  Developed a deep learning system to diagnose diseases from X-ray images using CNNs with TensorFlow, Keras, and PyTorch. Gathered and preprocessed labeled medical images with OpenCV, and used data augmentation to improve model accuracy. Assessed performance with metrics such as accuracy, precision, and recall with cross-validation. Deployed through Flask API and cloud platforms (AWS/GCP) for real-time diagnosis, offering a scalable solution to support radiologists and alleviate diagnostic workload.